Week 7 Notes

Review of Functions and Their Graphs:
Functions, Graphs, Tangents, Limits and Continuity

Course: Mathematics–1

Notes created by: Vishal (Instructor)

Lecture 1 — A Review of Functions and Geometry

Functions of one variable; linear, quadratic and polynomial functions; exponentials, logarithms, monotonicity and tangent lines

1 What this review covers

Before moving on to calculus, it is worth refreshing the ideas about functions and the geometry corresponding to them — the graphs. Once these basic concepts are in place, we can go on to capture the same ideas in terms of calculus.

Topics being reviewed:

- the straight line;
- quadratic equations;
- polynomials in general;
- functions — in particular, functions from real numbers to real numbers;
- exponentials and logarithms.

Other kinds of functions exist — for instance functions defined on the natural numbers — but those are not our concern here; later we shall study functions on $\mathbb{R}^n$.

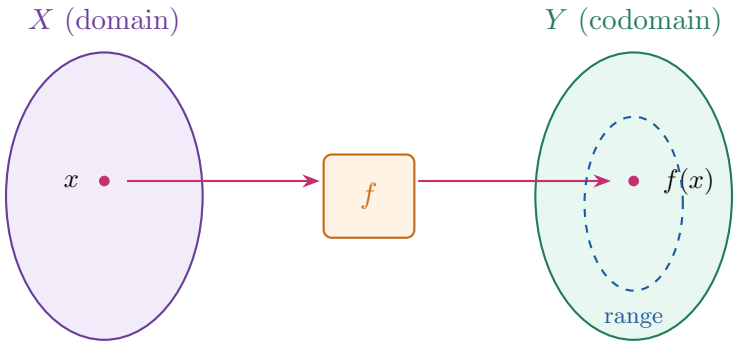
2 What is a function?

A function

A function is a **relation** from a set of inputs to a set of possible outputs, where each input is related to **exactly one** output.

This is a slightly cumbersome sentence — and it is more a heuristic than a formal definition — but it says the following. If the object x lies in the set of inputs, called the **domain**, then the function f will map, take, change, translate or transform the object x into exactly one object $f(x)$ in the set of possible outputs, called the **codomain**.

Writing $f : X \to Y$ means f is a function from X to Y, where X is the domain and Y is the codomain.



Range

The **range** of f is the set of values that f actually takes — that is, the subset of the codomain Y given by

$$\{\, y \in Y \;:\; y = f(x) \text{ for some } x \in X \,\}.$$

2.1 Function of one variable

For our purposes, a **function of one variable** means a function

$$f : D \longrightarrow \mathbb{R}, \qquad D \subseteq \mathbb{R},$$

where the domain D is a subset of the real numbers — not necessarily the whole of $\mathbb{R}$ — and the codomain is $\mathbb{R}$. We write it as $f(x)$, where x is a real number; hence *one* variable. If instead $D \subseteq \mathbb{R}^2$, we would call it a function of two variables, and so on.

3 Linear functions

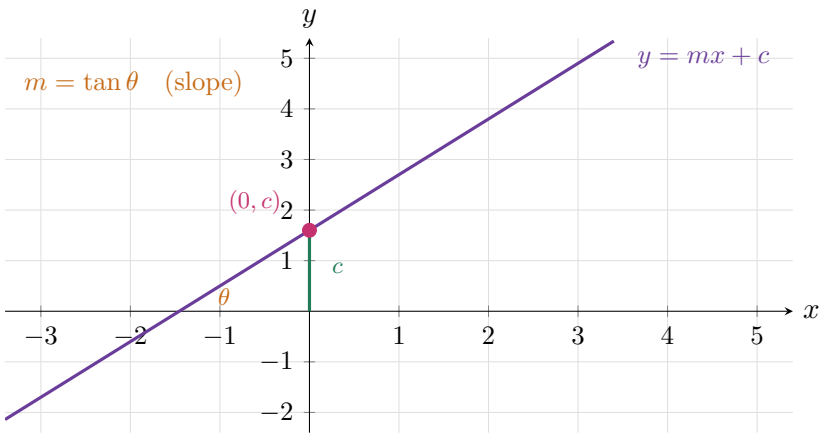
Linear function

$$f(x) = mx + c, \qquad m, c \in \mathbb{R}.$$

For every x this gives some real number. For example, with $f(x) = 5x + 2$:

$$f(0) = 5 \times 0 + 2 = 2, \qquad f(20) = 5 \times 20 + 2 = 100 + 2 = 102.$$

The associated picture is its **graph**: plot the point x and the corresponding value $f(x)$ on the y-axis, i.e. plot all points $\big(x, f(x)\big)$. In other words, the graph is the line $y = f(x)$. One reason to call it a *linear* function is precisely that the graph is a line.



Here m is the **slope** of the line, related to the angle θ at which the line is inclined to the x-axis by

$$m = \tan\theta,$$

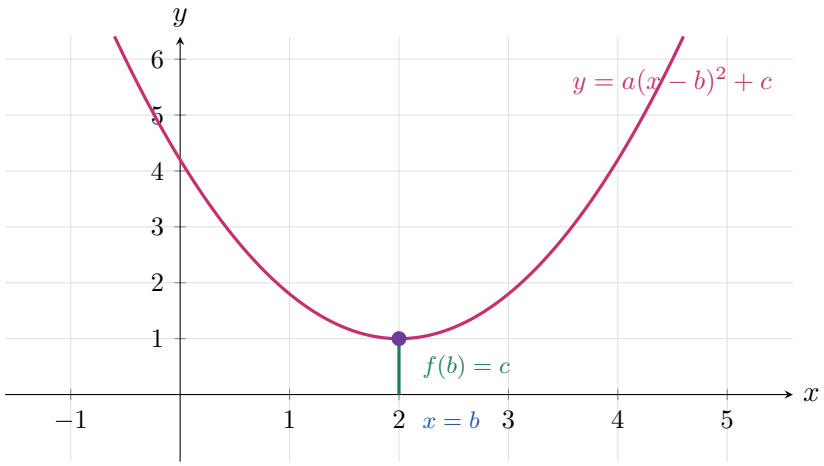
and c is the **y-intercept**: putting $x = 0$ gives $f(0) = c$, so the graph meets the y-axis at $(0, c)$ and c is exactly that length. Note that this "c" may be positive (on the positive side of the y-axis) or negative (on the negative side).

4 Quadratic functions

Quadratic function

$$f(x) = a\,(x - b)^2 + c.$$

The graph of such a function is called a **parabola**. Unlike a line, this is an example of a **curve** — something you can draw in a single stroke, without raising your pen, but which is not a line. (A line is a particular case of a curve, but when it is a line we simply say line.) Note also that graphs are particular kinds of curves: not every curve is a graph.



Reading a, b and c off the picture:

- If the parabola opens **upward**, then $a > 0$; if it opens **downward**, then $a < 0$.
- When $a > 0$, the smallest value of f is attained when $x = b$ — that is when the squared term contributes the least, namely 0. In the picture above, $b = 2$.
- Substituting $x = b$,
$$f(b) = a\,(b - b)^2 + c = c,$$
so the height of the lowest point is exactly c.

So a, b and c have everything to do with the geometry of the parabola, and we can travel in both directions: from the graph we can determine a, b, c; and from a, b, c we know the function and hence can draw the graph.

Algebra and geometry

There is an interplay here between the **algebra** — the functional form $a(x-b)^2 + c$ — and the **geometry** — the actual picture. Understanding how the two relate to each other is the goal of this part of the subject, and one way to do it is to study calculus.

4.1 Expanding the quadratic

Let us expand and see what we get:

$$a(x-b)^2 + c = a\left(x^2 - 2bx + b^2\right) + c = ax^2 - 2ab\,x + ab^2 + c.$$

Renaming the coefficients as $b' = -2ab$ and $c' = ab^2 + c$, this is

$$f(x) = ax^2 + b'x + c',$$

which is nothing but a **polynomial in x of degree 2**.

5 Polynomial functions

Polynomial function

A function $f : \mathbb{R} \to \mathbb{R}$ given by

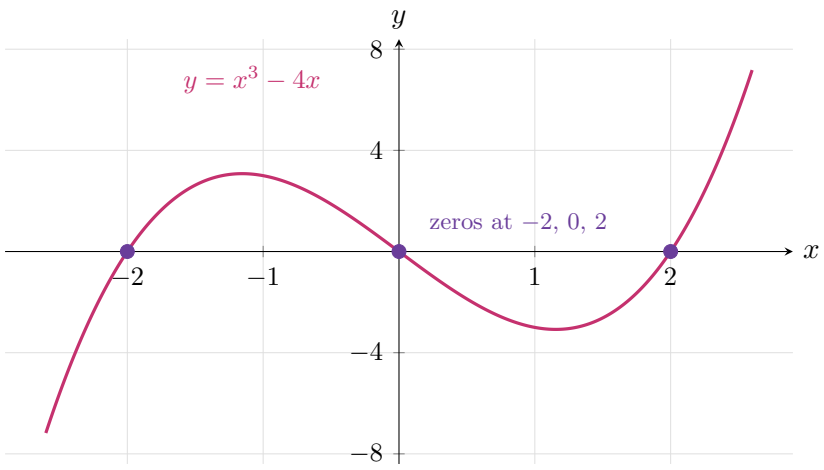
$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, a_n \neq 0$$

is a **polynomial of degree n**. The numbers $a_i \in \mathbb{R}$ are the **coefficients** of the polynomial.

Example 1.1

Study the function $f(x) = x^3 - 4x$ and find its zeros.

The graph is the set of points $\big(x, f(x)\big)$ as x runs over $\mathbb{R}$.



It is a nicely symmetric curve — not about the x- or y-axis, but symmetric about the origin. As x becomes large the function shoots off to infinity, and as x becomes small it shoots off towards minus infinity.

From the picture we can also see that there are three points at which the value becomes 0. Here is where the algebra kicks in: set $f(x) = 0$ and factor,

$$x^3 - 4x = 0 \implies x\left(x^2 - 4\right) = 0 \implies x\,(x-2)(x+2) = 0.$$

So $f(x) = 0$ exactly when

$$x = -2, \qquad x = 0, \qquad x = 2.$$

These are the zeros of the polynomial.The coefficients again play a role in the geometry of the graph — and this is one of the starting points of calculus.

6 Exponential functions

Let $f : \mathbb{R} \to \mathbb{R}$ be given by

$$f(x) = a^x.$$

What does a^x actually mean? This is a question in itself, and the answer is built up in stages.

6.1 Integer exponents

When the exponent is an integer the value is clear:

$$2^2 = 4, \qquad 0.5^2 = \frac{5}{10} \times \frac{5}{10} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = 0.25, \qquad \pi^2 = \pi \times \pi,$$

and the last may be approximated to as good a value as we want. Negative integers are equally clear:

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}, \qquad 0.5^{-2} = \frac{1}{0.5^2} = \frac{1}{\frac{1}{4}} = 4, \qquad \pi^{-2} = \frac{1}{\pi^2}.$$

6.2 Rational exponents

What if the exponent is not an integer? Consider

$$2^{1/3}.$$

The idea is that this is the **cube root of** 2: the number which, when raised to the third power, gives 2. That such a number exists needs proof, which we do not go into here.

On similar lines,

$$0.5^{5/6}$$

is understood by first computing 0.5^5 and then taking its sixth root — it is that number whose sixth power equals 0.5^5. In general, for $x = \frac{m}{n}$ with m, n integers,

$$a^{m/n} = \sqrt[n]{a^m}.$$

6.3 Irrational exponents

Now the function $f(x) = a^x$ has x running over *all* real numbers, so we must also deal with irrational exponents such as

$$2^{\sqrt{2}} \qquad \text{and} \qquad 2^{\pi}.$$

How an irrational exponent is defined

Take a **rational number** q which approximates $\sqrt{2}$ as closely as we desire — looking up $\sqrt{2}$ in a table gives 1.something, which can be written as a fraction such as $\frac{\text{something}}{100}$. We know how to compute 2^q from the rational case above, and $2^{\sqrt{2}}$ is very closely approximated by 2^q. Choosing a rational closer to $\sqrt{2}$ gives a better approximation — and that is how quantities like $2^{\sqrt{2}}$, and even 2^{π}, are defined.

Why a must be positive

The base a had better be a positive number: if a were negative we would struggle badly when taking roots, and if a were 0 we would run into trouble on dividing. So for a^x to make sense on the entire domain $\mathbb{R}$, we require

$$a > 0.$$

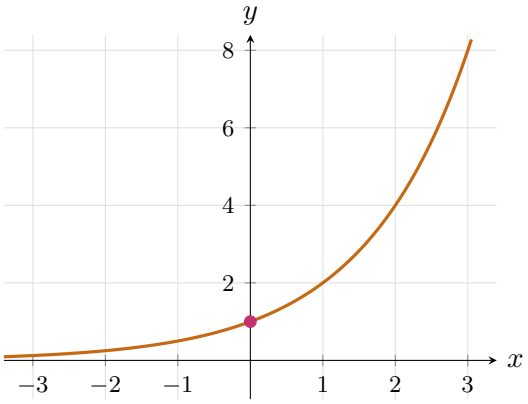
7 Logarithmic functions

The logarithm is the **reverse process**: when you reverse out the exponential function, you get the logarithm. Since a^x takes only positive values (with $a > 0$), the logarithm accepts only positive inputs and returns values across the whole real line. The relation between the two, at base 2, is

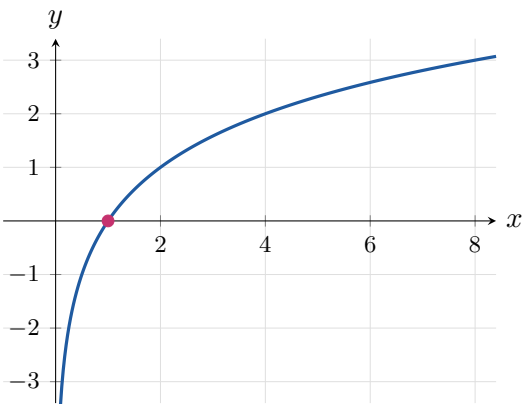
$$\log_2 (2^x) = x,$$

which is what is meant by *composing* the exponential and the logarithm.

$y = 2^x$: grows very rapidly



$y = \log_2 x$: increases very slowly



Both functions are increasing, but they play exactly opposite roles: the exponential grows very rapidly while the logarithm increases very slowly.

8 Monotone functions

Monotone increasing and decreasing

A function f is **monotone increasing** if

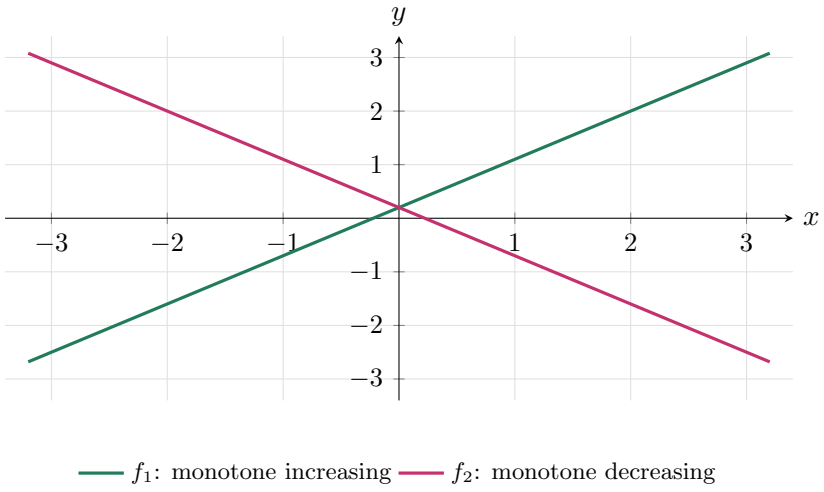
$$x_1 \leq x_2 \implies f(x_1) \leq f(x_2),$$

and **monotone decreasing** if the opposite happens:

$$x_1 \leq x_2 \implies f(x_1) \geq f(x_2).$$

(One also says *monotonically* increasing or decreasing — they mean the same thing.)

In words: for an increasing function, as x increases the value $f(x)$ increases; for a decreasing function, as x increases the value $f(x)$ decreases.

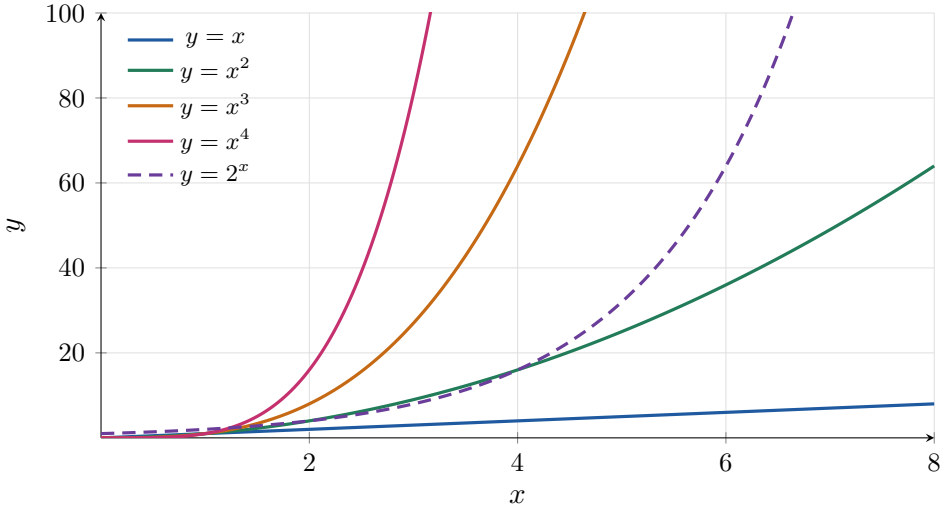


These definitions are worth keeping in mind, because eventually we study maximum and minimum values, and for that we need to know where functions are increasing and where they are decreasing.

9 Comparing growth rates

9.1 Powers against the exponential

Compare the monomials $y = x$, $y = x^2$, $y = x^3$, $y = x^4$ with the exponential $y = 2^x$.



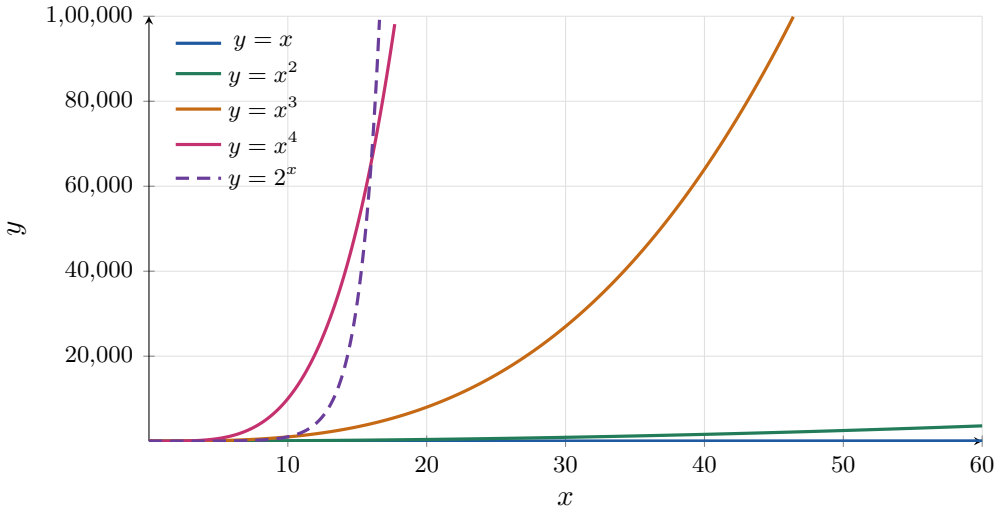
Reading the picture:

- Up to $x = 1$, the line $y = x$ is actually the largest of the monomials — the others lie below it. At $x = 1$ they all become equal to 1.
- After that, x^2 starts going faster than x, x^3 even faster, and x^4 faster still: as the exponent grows, the monomial grows more rapidly.
- The exponential 2^x starts above 0, is then caught by almost all the others (though never by $y = x$, which it always beats), and after a short while it takes off with respect to x^2 and keeps changing.

A word about the axes

Be careful with the labelling of the axes in such comparison pictures. If both axes were drawn on the same scale, $y = x$ would appear at 45°; but since we want many more values on the y-side, that axis is shrunk, and so $y = x$ does not appear at 45°.

Now increase the scale — let x run up to 60 and y up to $10^5 = 1\,00\,000$:



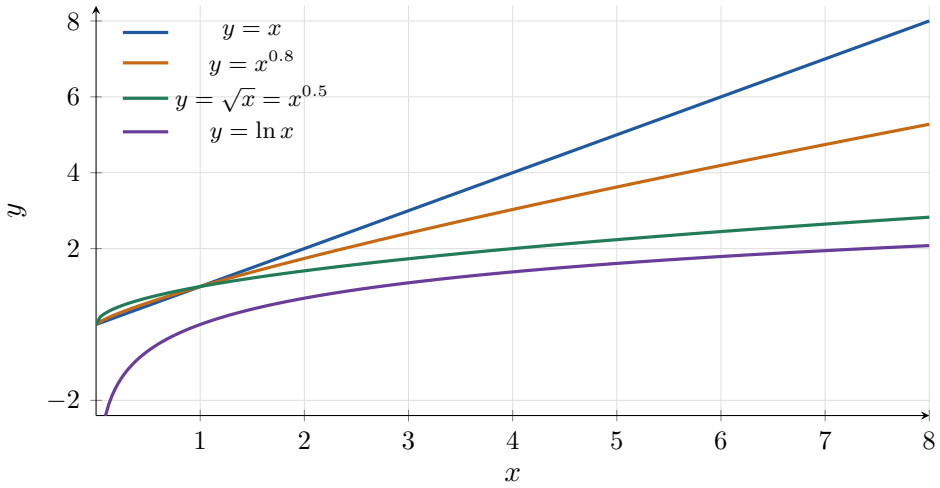
On this scale $y = x$ almost looks like a flat line, and even $y = x^2$ looks somewhat flat (it is not — drawn properly it is our familiar parabola). The cubic is fairly fast even here, and $y = x^4$ faster; but $y = 2^x$ tilts further still and beats x^4 as well.

The point of these pictures

As n increases, the functions $y = x^n$ go off to infinity more and more rapidly — but **all of them are eventually overtaken by the exponential function**. After some point the exponential always beats any polynomial. (This is intuition rather than a formal requirement of the course, but it is good to know how these behave.)

9.2 Slow-growing functions

On the other side we have the slow growers: $y = x$, $y = x^{0.8}$, $y = \sqrt{x} = x^{0.5}$, and the logarithm.



Here $y = x$ is a line, while $y = x^{0.8}$ is a very slow-growing curve that may look somewhat linear; $\sqrt{x}$ lies below it. The logarithm is defined only on the positive side, starts very far below, comes up, then changes direction and proceeds very slowly. The same phenomenon as before occurs in the opposite direction:

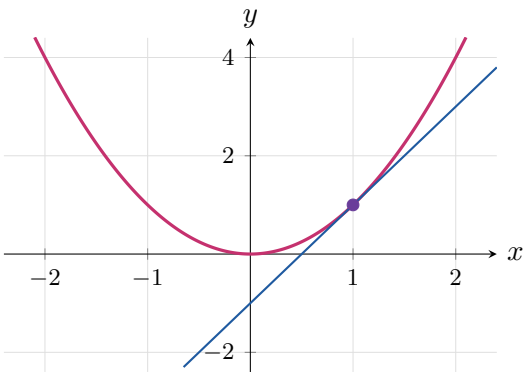
Slow growth

The **logarithm grows more slowly than any polynomial**.

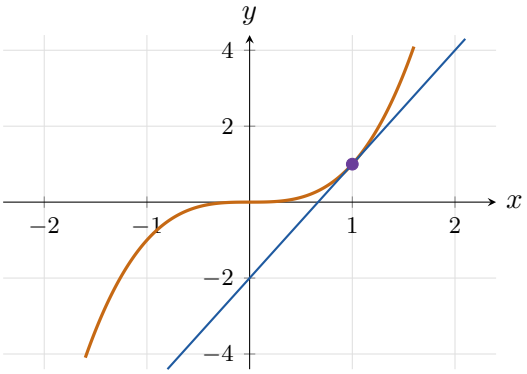
10 Tangent lines

Finally, recall the idea of a **tangent line**. Take a line that intersects the curve you have. It may intersect at many points; now move it slowly, parallel to itself, until the points of intersection all come together into a single point. The line obtained in this way is called a **tangent line**. (This is only a first description; we make it precise — and see why "touches at exactly one point" is misleading — in Lecture 3.)

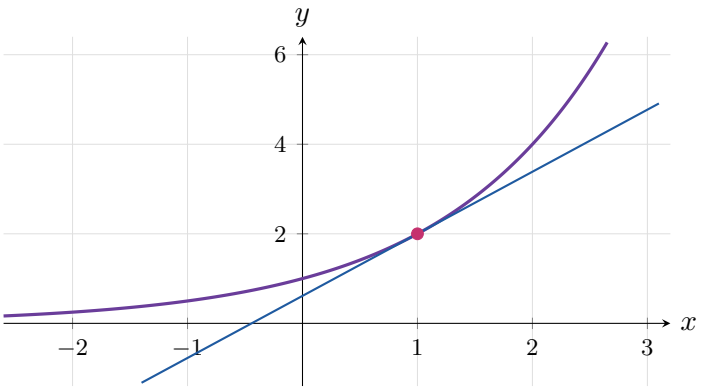
$y = x^2$ with tangent at $x = 1$



$y = x^3$ with tangent at $x = 1$



$y = 2^x$ with tangent at $x = 1$



Such a tangent line, being a line, is given by

$$y = mx + c,$$

and the natural question is: how do we find the equation of that line? That is exactly what calculus will let us do.

Lecture 2 — Functions of One Variable

Trigonometric functions and identities; arithmetic operations on functions; composition

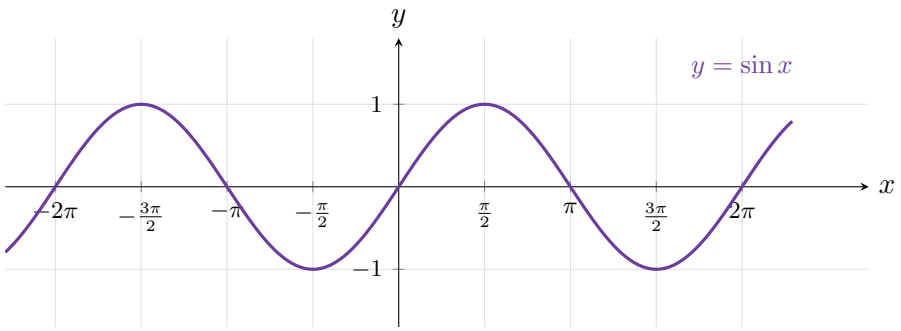
1 Trigonometric functions

Recall that a function of one variable is a function from some subset of $\mathbb{R}$ to $\mathbb{R}$. The common examples we have all *heard* of — though perhaps not studied as functions — are the **trigonometric functions**. We treat them here.

1.1 The sine function

The sine function is

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = \sin x.$$



It is a very nice graph: it is **periodic** — such graphs are called periodic or **sinusoidal** — it varies between -1 and 1, and it has **period** 2π. Looking at its trajectory between -2π and 0, it starts, rises, falls, and then restarts; shifting it by 2π does not change it at all. The graph is the set of points $\big(x, f(x)\big)$ with $x \in \mathbb{R}$, and of course it keeps going over the entire real line.

How the value is computed

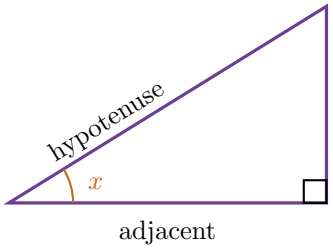
Given a real number x, how do we actually compute $\sin x$? The following is a heuristic description — there is a more formal technical definition, which we avoid here.

Building $\sin x$ in stages

1. For x between 0 and $\frac{\pi}{2}$, use the right-angled triangle:

$$\sin x = \frac{\text{length of the opposite side}}{\text{length of the hypotenuse}}.$$

2. For x between $\frac{\pi}{2}$ and π, use the **symmetry** of the picture: the value at a point is read off from the corresponding point in the first stretch. This gives sin on $(0, \pi)$.
3. For x between π and 2π, **reflect** (or, if you prefer, rotate) the picture, which gives sin on the whole of $(0, 2\pi)$.
4. For any other x, add or subtract suitable multiples of 2π so as to bring it inside $(0, 2\pi)$, and use the value there.



$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos x = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan x = \frac{\text{opposite}}{\text{adjacent}}$$

The ratio does not depend on the triangle

What we are exploiting is that such a ratio is **constant** — it depends only on the angle, not on how large the triangle is. Drawing a bigger right-angled triangle with the same angle x leaves the ratio unchanged, which can be proved by the similarity of triangles.

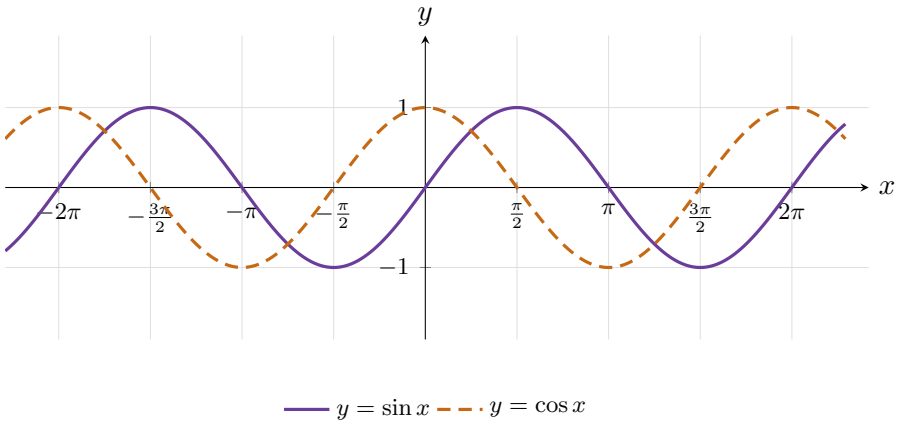
1.2 The cosine function

The idea is the same. For x between 0 and $\frac{\pi}{2}$,

$$\cos x = \frac{\text{length of the adjacent side}}{\text{length of the hypotenuse}},$$

and beyond that the picture shows how to extend it, because of the periodic nature of the function.

The cosine has properties similar to the sine: it is **periodic with period** 2π, it is sinusoidal, and it takes values between -1 and 1. That last part is clear, since the adjacent side cannot be longer than the hypotenuse, so the value is at most 1; the minus sign is allowed because the picture is flipped below — the sign keeps track of which quadrant we are in.



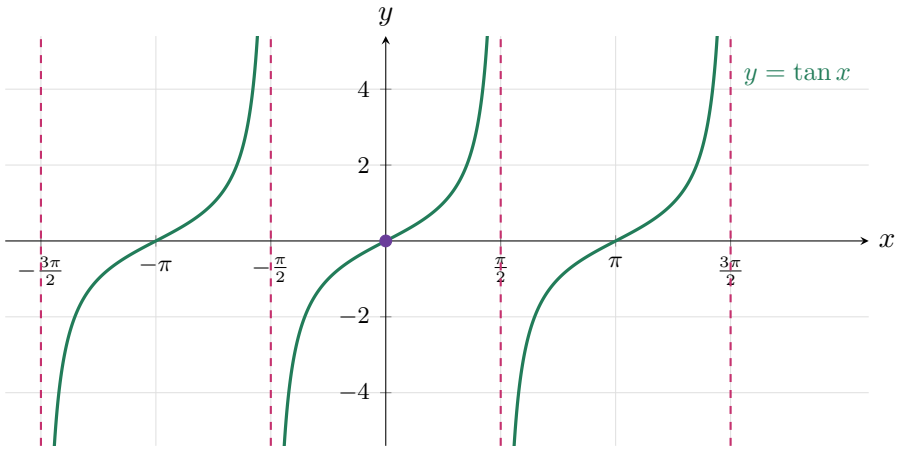
Put together, the two make a very pleasant picture, and one sees at once that each is simply a **shift of the other**: shifting by $\frac{\pi}{2}$ turns one into the other.

1.3 The tangent function

Next is the tangent function, $f(x) = \tan x$, for which a caveat is needed — it is *not* defined on the whole of $\mathbb{R}$. For x between 0 and $\frac{\pi}{2}$,

$$\tan x = \frac{\text{length of the opposite side}}{\text{length of the adjacent side}},$$

and the symmetry of the function then gives its values from $-\frac{\pi}{2}$ to 0.



Reading the graph: between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ the curve rises very rapidly, then becomes relatively slower, passes through $(0,0)$, and shows the opposite behaviour on the other side before taking off again. At the red dashed lines — which are $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$ — the function runs off to $-\infty$ and $+\infty$. Since we are working with real numbers, we regard the function as **undefined** there.

Domain of the tangent

$$\tan : \mathbb{R} \setminus \left\{ \frac{n\pi}{2} \ : \ n \text{ odd} \right\} \longrightarrow \mathbb{R}$$

So we never ask for $\tan\left(-\frac{\pi}{2}\right)$, $\tan\left(\frac{\pi}{2}\right)$, $\tan\left(-\frac{5\pi}{2}\right)$ or $\tan\left(\frac{7\pi}{2}\right)$ — at such numbers the tangent is not defined. Note also that tan is periodic with period π (not 2π), so a shift by π each time gives the remaining values.

Why the name?

The tangent function ought to have something to do with the tangent line we met in the previous lecture — and indeed it does. That connection is something we will explore using calculus.

2 The remaining trigonometric functions and some identities

Besides these three we have the **cotangent**, **secant** and **cosecant**:

$$\cot x = \frac{1}{\tan x}, \qquad \sec x = \frac{1}{\cos x}, \qquad \operatorname{cosec} x = \frac{1}{\sin x}.$$

The last two may be taken as definitions. In every case one must be careful about the domain, since not all of $\mathbb{R}$ will work — the points where $\cos x = 0$ or $\sin x = 0$ must be thrown out.

Some identities

$$\sin(-x) = -\sin x$$
$$\cos(-x) = \cos x$$
$$\tan(-x) = -\tan x$$

$$\sin^2 x + \cos^2 x = 1$$
$$\tan x = \frac{\sin x}{\cos x}$$
$$\sec x = \frac{1}{\cos x}, \qquad \operatorname{cosec} x = \frac{1}{\sin x}$$

A few remarks:

- The first three identities are exactly how we extended the functions beyond $\left(0, \frac{\pi}{2}\right)$.
- $\sin^2 x + \cos^2 x = 1$ is a very important identity, and a straightforward consequence of the so-called Pythagoras theorem — which was, of course, known in many civilisations long before Pythagoras.
- In $\tan x = \dfrac{\sin x}{\cos x}$ there is a caveat: the denominator must be non-zero. And $\cos x = 0$ exactly at the points $\frac{n\pi}{2}$ with n odd — precisely the points we already ruled out when defining the tangent.
- There are further identities we do not write out: for instance $\sin\left(x + \frac{\pi}{2}\right)$ can be expressed in terms of $\cos x$, and $\cos\left(x + \frac{\pi}{2}\right)$ in terms of $\sin x$. The graph itself tells you what those identities must be.

3 Arithmetic operations on functions

Let us now turn to some general properties of functions of one variable. We would like to add, subtract, multiply and divide functions — and indeed we can do so without any problem.

The four operations

Let $D \subseteq \mathbb{R}$ and let f, g both have domain D. Then, for $x \in D$:

Sum:	$(f+g)(x) = f(x) + g(x)$
Product:	$(fg)(x) = f(x)\, g(x)$
Scalar multiple:	$(cf)(x) = c\, f(x), \qquad c \in \mathbb{R}$
Quotient:	$\left(\dfrac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}, \qquad$ where $g(x) \neq 0$

For example, $\sin + \cos$ evaluated at x is simply $\sin x + \cos x$. We have already met the idea of a sum when constructing polynomials: monomials are summed, with scalar multiplication supplying the coefficients. Note also that the product is often written $f(x)\, g(x)$ with the times sign removed, and the scalar multiple as $cf(x)$.

The quotient may shrink the domain

Sometimes f and g are both defined on all of D — sin and cos are defined on the whole real line — yet there are values at which g vanishes. Dividing sin by cos therefore forces us to restrict the domain to those values where $\cos x \neq 0$. That is exactly why

$$\tan x = \frac{\sin x}{\cos x}$$

is defined on all of $\mathbb{R}$ *except* the odd multiples of $\frac{\pi}{2}$. So the quotient need not be defined on all of D.

3.1 Building rational functions

The fourth operation lets us create **rational functions**.

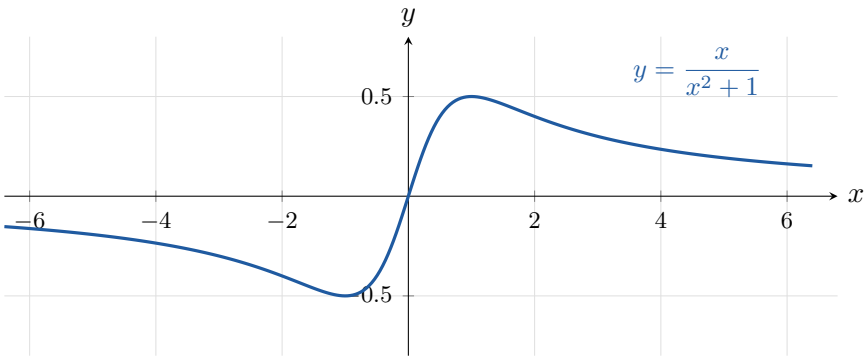
Example 2.1

Consider

$$h(x) = \frac{x}{x^2 + 1}.$$

Here x^2+1 is never 0, because a square plus one is always greater than or equal to 1. So this function makes sense as a function from $\mathbb{R}$ to $\mathbb{R}$. We obtained it by taking

$$f(x)=x, \qquad g(x)=x^2+1, \qquad h=\frac{f}{g}.$$



4 Composition of functions

Composition

Let $D\subseteq\mathbb{R}$ and let f be a function on D; let $g:E\to\mathbb{R}$, where the **range of f lies inside E**. Then for each $x\in D$ we have $f(x)\in E$, so $g\big(f(x)\big)$ makes sense and yields a well-defined real number. This gives a function

$$g\circ f: D\to\mathbb{R}, \qquad (g\circ f)(x)=g\big(f(x)\big),$$

called the **composition** of f and g.

The condition that the range of f lies in E is essential.

Example 2.2

Let

$$f(x)=x^2+1 \quad (f:\mathbb{R}\to\mathbb{R}), \qquad g(x)=\sqrt{x} \quad \big(g:[0,\infty)\to\mathbb{R}\big).$$

Find $g\circ f$ and check that the composition makes sense.

Note first that the domain of g is restricted — the square root needs a non-negative argument. The composition is

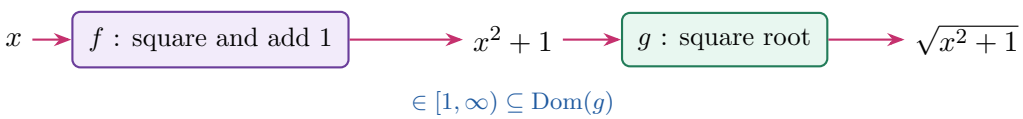
$$(g\circ f)(x)=\sqrt{x^2+1}.$$

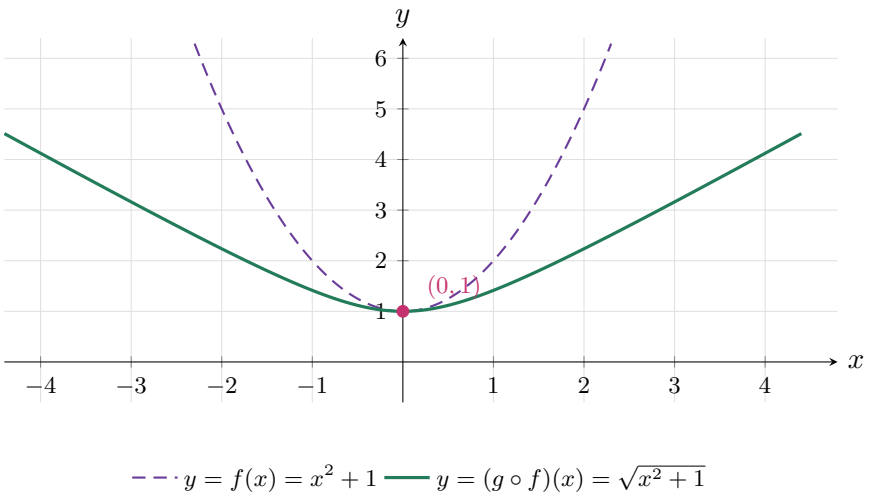
Why may we talk about $g\circ f$ at all? Look at the range of f: it consists of all values y such that $y=x^2+1$ for some x. The smallest value taken is 1, and every value above it is taken, so

$$\mathrm{Range}(f)=[1,\infty).$$

This contains no negative numbers, which is exactly why taking the square root makes sense:

$$[1,\infty)\subseteq[0,\infty)=\mathrm{Dom}(g).$$





So: functions of one variable can be added, scalar multiplied, multiplied, and — provided the denominator is non-zero — divided by one another, as long as they share a domain; and when the range of one matches nicely into the domain of the other, they can be composed.

Lecture 3 — Graphs and Tangents

The graph as a set; what a curve is; tangent lines to curves and to functions

1 The graph of a function

Definition 3.1 — Graph

If $f : X \to Y$ is a function, the **graph** of f is the subset

$$\Gamma_f = \{\, (x,\, f(x)) \;:\; x \in X \,\} \;\subseteq\; X \times Y.$$

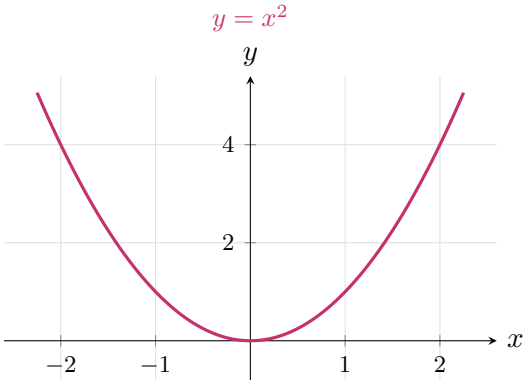
This is the formal definition, and note what it says: the graph of a function is a **set**. In principle it has nothing to do with geometry or calculus. But typically the sets we have in mind are $X, Y \subseteq \mathbb{R}^n$ — very often simply $\subseteq \mathbb{R}$, in which case f is a function of one variable — and then we can give Γ_f a geometric meaning by plotting it in $\mathbb{R}^2$.

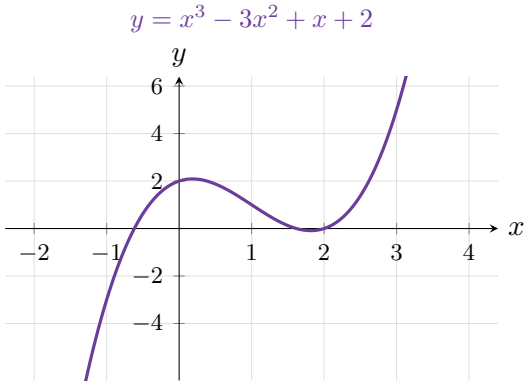
So if $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$, the graph of f can be drawn as a curve in $\mathbb{R}^2$ by considering the points

$$\{\, (x, y) \;:\; y = f(x) \,\}.$$

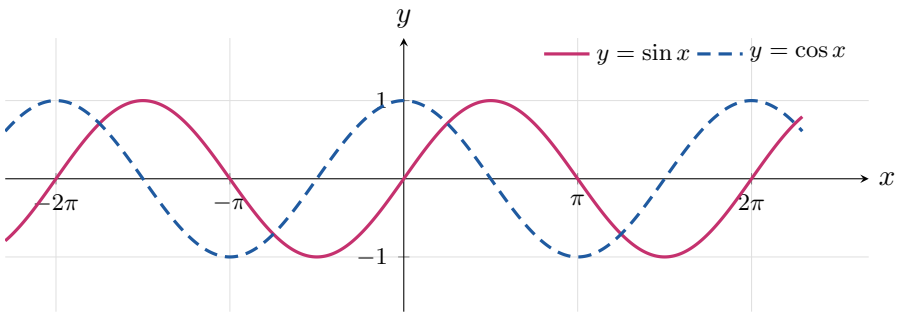
Here D is a priori just any subset of $\mathbb{R}$ — very often the entire real line, but for some functions we must restrict D to where the function is actually defined, and we have seen such examples before.

1.1 A gallery of graphs

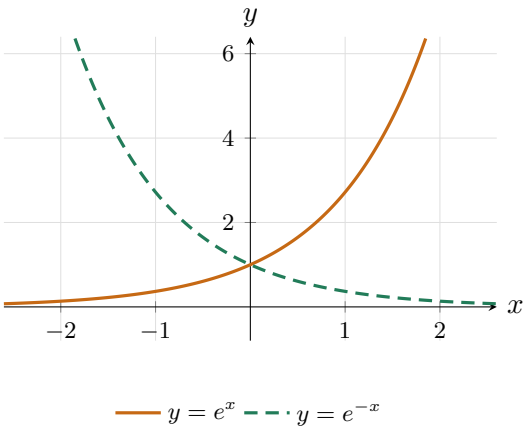


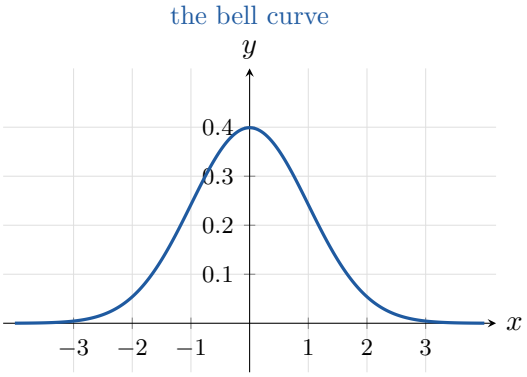


For the cubic on the right: between roughly $x = -1$ and $x = 3$ the graph is fairly bounded and does not increase too fast; but beyond -1 and beyond 3 it rises very steeply. As x increases it is the x^3 term that controls how fast the function grows — the same idea we met when comparing rates of increase.



These are the **periodic** or **oscillating** functions, and the reason for the name is clear right from the graph. In each case the curve is the set $\{(x, \sin x)\}$ or $\{(x, \cos x)\}$ as x runs over the whole of $\mathbb{R}$.





The function $y = e^{-x}$ is what is called **exponential decay**, a term you may have come across before. The second picture is the graph of

$$y = \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2},$$

often called the **bell curve** because it looks like a bell; it corresponds to the **normal distribution** in statistics.

In every case the graph is a **curve**, and it is simply the set of points $\big(x, f(x)\big)$.

2 What is a curve?

We keep saying "this is a curve". What is a curve? The precise mathematical definition is more complicated than one might think, so here is a heuristic picture — which most of us already carry.

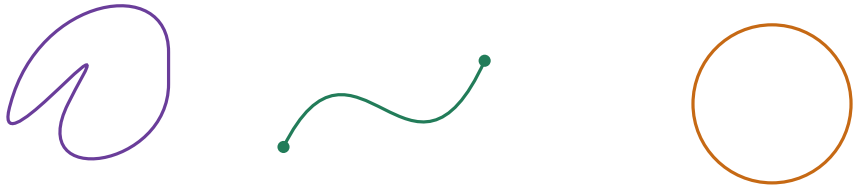
A curve (heuristically)

A **curve** is a figure obtained as the **path of a moving point**.

Think of an aeroplane flying, a bee flying, the motion of a cricket ball when it is bowled or when it is hit by the batsman, or the movement of a figure skater. Whatever motion you can think of, the path traced by that motion is a curve.

- A motorcycle moving in a straight line with uniform speed traces a **line** — perhaps the line $y = x$. So **straight lines are also curves**.
- A plane flying makes an arc and then goes away; in an air show it may go round in a loop that crosses itself. Drawn in $\mathbb{R}^2$ the path appears to intersect itself, but if you think of it in $\mathbb{R}^3$ the two strands are actually separate — the apparent crossing comes from projecting into the plane.

Another way of thinking about it: a curve is a figure obtained by **bending a line** at various places. Sometimes the bends are very steep. And we allow the line to come back and meet itself.



a closed curve an open curve (two ends) a circle

The first is a **closed curve**: it comes back and meets itself. The second is not closed — it has two ends, so you may think of it as a piece of string. The third is the most familiar of all, and has nice geometric properties, but note that when we say *curve* we do not require anything so tidy: a strange-looking shape or the trajectory of an aeroplane is equally a curve.

3 The tangent line to a curve

Definition 3.2 — Tangent line

The **tangent line** to a curve C at a point p (which must lie on C) is the line representing the **instantaneous direction** in which the curve C moves at the point p.

That sounds complicated, so here is what it means. Think of the curve as the motion of a particle. When the particle is at the point p, consider the instant just before and the instant just after. In that time, in what direction is it moving? The direction is determined by an angle, and the tangent line is the line through p in that direction. The emphasis is on the word **instantaneous**.

The traditional description

Traditionally — and this is how many elementary textbooks put it — the tangent is described as "a line which just touches the curve at that point". It is never made clear what *just touches* means, so we understand it through examples instead.

A common misconception

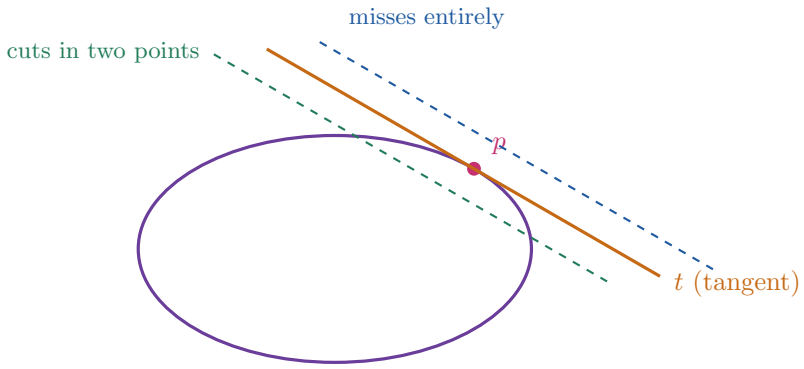
It is **not** necessary that the tangent line intersects the curve only once. For a general curve the tangent at p may well cut the curve again elsewhere. For a circle it happens to touch at exactly one point, but that is a special case — this mistake is often propagated, so be warned.

3.1 A frequently-seen property

Often **— but not always**

A tangent line to a curve C at a point p *often* has the property that it passes through p but does not intersect C at any other point **close to** p; and lines parallel and close to it either do not intersect C close to p, or intersect C in two or more points close to p.

The picture explains it better than the words. Take an ellipse and a point p on it, with tangent line t. Then t does not meet the ellipse anywhere near p — in fact not at any other point at all. But shift t slightly, parallel to itself: either it misses the ellipse entirely, or it cuts it in two points.



The same phenomenon shows up elsewhere. For a circle, a parallel line either cuts it in two points or misses it altogether — no surprise, since a circle is a special ellipse.

Why "often" is in quotes

This is **not** a definition and not a general phenomenon. Here is the cleanest counterexample: a **line** is itself a curve. What is the tangent line to a line? It is the same line. But now move it slightly — it does not intersect the original line at all on either side; and the tangent itself intersects the curve *everywhere*, not just at p. So the idea that a tangent line meets the curve only at that one point is simply not correct.

4 The tangent line to a function

Definition 3.3 — Tangent line to a function

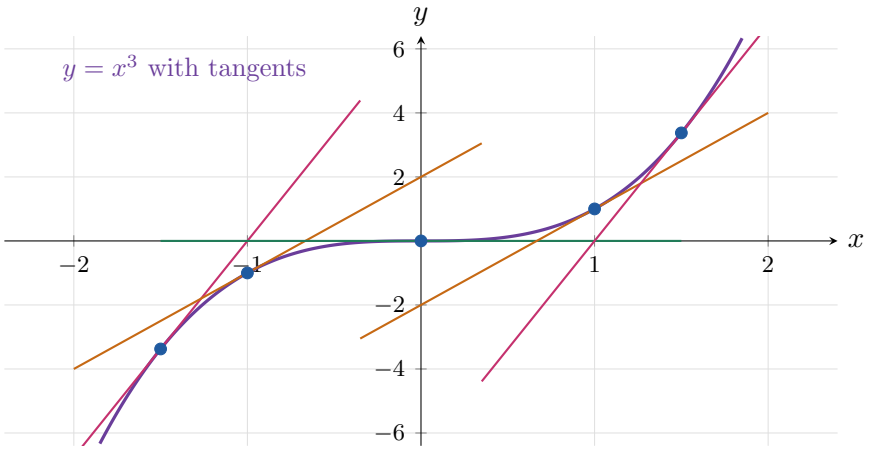
Let f be a function and assume Γ_f is a curve. Let x be a point of the domain at which $f(x)$ is defined. Then the **tangent line to f at x** is the tangent line to the graph of f at the point $\big(x, f(x)\big)$.

Note the assumption: the graph Γ_f need not always be a curve, and we shall see examples where it is not one in the sense described above.

Example 3.1

Take $f(x) = x^3$ and draw tangent lines at several points.

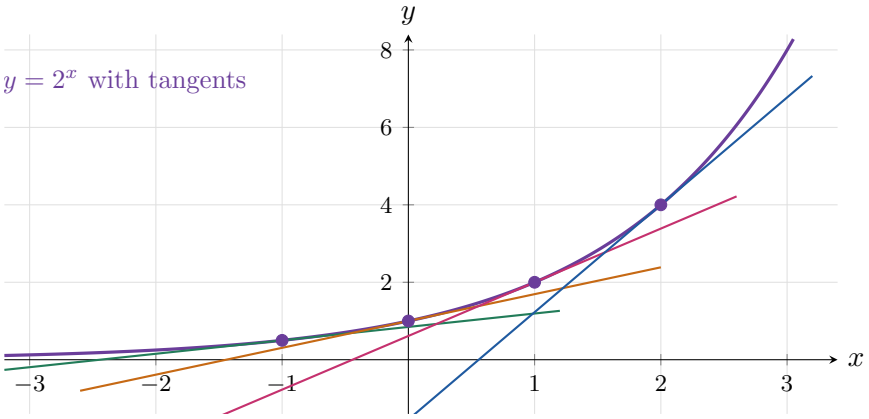
To find the tangent to f at the point 0, look at 0 on the axis and then at the corresponding point on the graph — here $(0, 0)$. For any other point x, look at $\big(x, f(x)\big)$ and ask for the tangent to the graph there.



Notice how the tangent lines move in a very orderly way and keep changing. This picture really reflects the role of the tangent line: it tells you the direction in which the curve is moving, so if you draw *enough* tangent lines you can see the curve itself emerge from them.

Example 3.2

The same for $f(x) = 2^x$.



Again the tangent lines describe the graph extremely well: put all of them together and you can make out the trajectory of the particle, which is the curve.

So, for instance, to ask for the tangent line to $f(x) = \sin x$ at the point 0, we look at the curve $\Gamma_{\sin} = \{(x, \sin x)\}$, locate the point $\big(0, \sin 0\big) = (0, 0)$ on it, and ask for the tangent line to that curve at that point.

5 Two cautionary examples

5.1 The floor function

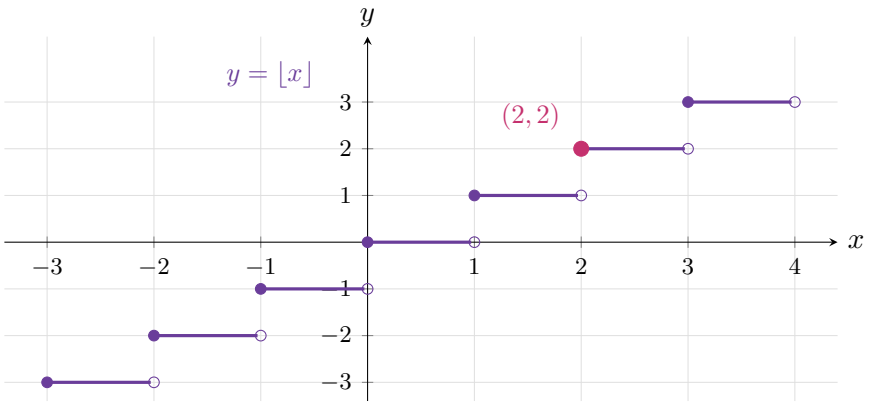
Example 3.3

For the **floor function** $\lfloor x \rfloor$, we take the largest integer less than or equal to the number:

$$\lfloor 1000.23 \rfloor = 1000, \quad \lfloor 6.829 \rfloor = 6, \quad \lfloor \pi \rfloor = 3, \quad \lfloor -2.896 \rfloor = -3, \quad \lfloor -20 \rfloor = -20.$$

What is the tangent line to this function at the point 2?

The graph is a **step function**: constant until the next integer, then a jump up by 1, constant again, and so on.



To find the tangent at $x = 2$, look at the corresponding point $(2, 2)$ and ask how the particle is moving instantaneously there. And now we are in big trouble, because there is a **huge jump** at that point: approaching 2 from below we are on one horizontal level, and just after it we are on another. There is no line that captures what is happening at that point.

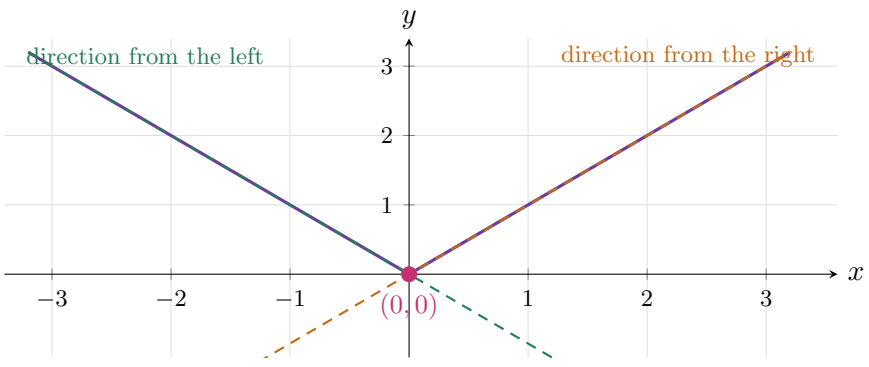
There is no tangent line to $y = \lfloor x \rfloor$ at $x = 2$.

5.2 The modulus function

Example 3.4

What happens for $y = |x|$ at the point 0?

Here we do not have the serious problem of the floor function — there are no jumps, so it is what is called **continuous**: you can draw it nicely with a pen without lifting it.



But recall the definition: the tangent should reflect the instantaneous direction of motion. As the particle comes close to $(0, 0)$ *from the left*, that direction is one line; thinking about what happens *after* it has left the point, it is a different line. One gives the instantaneous direction on one side and the other on the other side, and they do not match up in any way.

There is no tangent line to $y = |x|$ at $x = 0$.

These examples show that the idea of a tangent line is trickier than it seems when things are nice and smooth. If there are **jumps** or sudden **zigzags**, tangent lines may not exist at all. So we must be careful, and we need to understand the notion in a more conceptual — preferably more algebraic — way, by writing down equations.

For this to be properly understood and theorised took the invention of calculus, and then a considerable amount of time more. The next few lectures are devoted to developing exactly that theory: we will talk about **limits**, then **continuity**, and then the notion of a **derivative**.

Lecture 4 — Limits for Sequences

Sequences and their limits; convergence and divergence; subsequences; rules for computing limits

1 Why we need this

In the previous lecture we saw that the tangent is a tricky question — what exactly is it, and when does it exist? The examples at the end showed that there is more depth to the question than geometry alone reveals. To get a proper handle on it we must develop a little theory, and the route runs through the notion of **differentiability**. We begin with **limits for sequences**.

2 What is a sequence?

Sequence

A **sequence** is an assignment: for each number n there is some corresponding number, which we call a_n.

Examples of sequences:

$$1,\ 2,\ 3,\ 4,\ 5,\ 6,\ \ldots \qquad -1,\ 1,\ -1,\ 1,\ \ldots\ =\ (-1)^n$$

$$\frac{1}{2},\ \frac{1}{4},\ \frac{1}{8},\ \frac{1}{16},\ \frac{1}{32},\ \frac{1}{64},\ \cdots\ =\ \frac{1}{2^n}$$

Of course a sequence need *not* have a closed form at all: for each n you simply associate some number — that is a sequence.

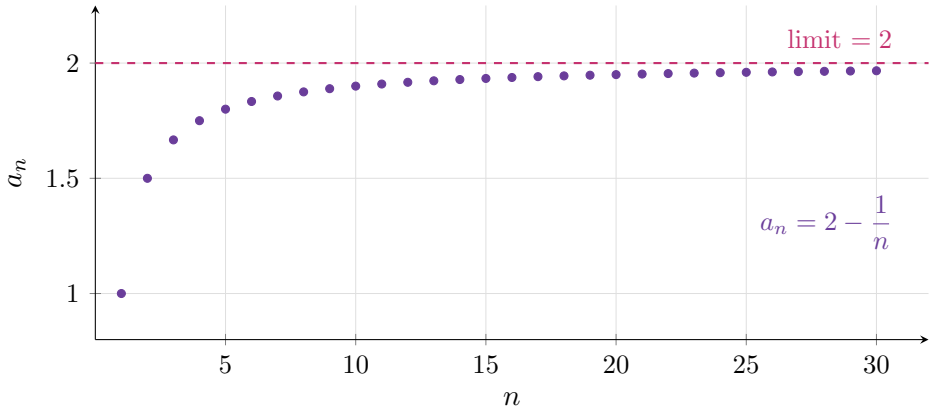
3 Three motivating sequences

3.1 The sequence $a_n = 2 - \frac{1}{n}$

What happens as n increases?

n	$2-\frac{1}{n}$
1	1
2	1.5
3	1.66…
⋮	⋮
130	very close to 2
200	1.995

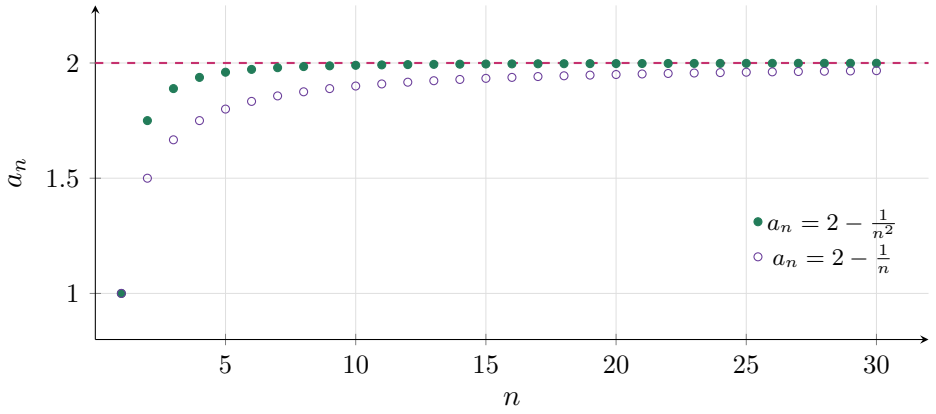
The first term is $2 - \frac{1}{1} = 1$; at $n = 2$ we get $2 - \frac{1}{2} = 1.5$, then $2 - \frac{1}{3} = 1.66\ldots$, and as n grows the value creeps up: 1.98, 1.988, 1.989, 1.99, 1.993, 1.994. Plotting the terms on the real line, the moving dot comes closer and closer to 2. By $n = 200$ we are at 1.995 — really close.



So as n becomes very large, this sequence comes very close to the number 2.

3.2 The sequence $a_n = 2 - \frac{1}{n^2}$

Now let us watch the *speed*. The first term is $2 - \frac{1}{1^2} = 1$; then $2 - \frac{1}{4} = 1.75$; then $2 - \frac{1}{9} = 1.8889$ — already at the third term, which took the previous sequence far longer. A few terms later it is 1.97, at $n = 30$ it is 1.999, and by $n = 50$ it is 1.9996.



On the real line the dot appears to sit at 2 already — but it does not; it is very close, and not exactly 2.

3.3 A slow sequence

Example 4.1

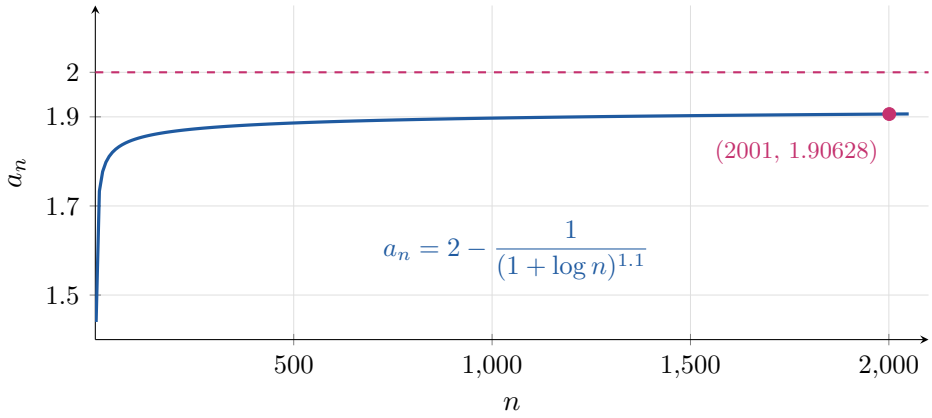
Consider

$$a_n = 2 - \frac{1}{(1 + \log n)^{1.1}}.$$

This one moves very slowly. Past $n = 300$ it has still not crossed 1.9; at $n = 367$ you may begin to suspect it will stop short of 2 altogether. Past 500 it is closing on 1.9; near $n = 1400$ it has just crossed 1.9 and is creeping along at $1.90\ldots$. Stopping at the 2001st term,

$$a_{2001} = 2 - \frac{1}{(1 + \log 2001)^{1.1}} = 1.90628.$$

Nevertheless the claim is that as n becomes very large this sequence does come very close to 2 — it just gets there very slowly.



Speed of convergence

All three sequences approach 2, but at very different speeds:

- $2 - \frac{1}{n}$: within 200 terms we reached 1.995 — reasonable.
- $2 - \frac{1}{n^2}$: within 50 terms we reached 1.9996 — fairly fast.
- $2 - \frac{1}{(1+\log n)^{1.1}}$: even at the 2001st term we had only 1.906 — relatively slow.

So besides *whether* a sequence approaches a limit, there is the question of *how fast* it does so.

4 The limit of a sequence

Definition 4.1 — Limit of a sequence

Let a_n be a sequence of real numbers. We say that a_n has **limit** a (some real number) if, as n increases, the numbers a_n come closer and closer to a.

It is natural to ask what "closer and closer" means precisely. There is a formal, abstract mathematical definition, which we do not get into here — we shall see just enough to make out what is going on.

4.1 Equivalent notations

All of the following mean exactly the same thing — that the sequence a_n has limit a:

- a_n tends to a
- a_n converges to a
- $\lim a_n = a$
- $a_n \to a$
- $a_n \xrightarrow[n\to\infty]{} a$
- $\lim_{n\to\infty} a_n = a$

Whatever the notation, the content is the same: **as n becomes larger and larger, the numbers a_n come closer and closer to a.**

5 Convergent and divergent sequences

Convergent, divergent

A sequence a_n is called **convergent** if it converges to some limit (a real number). A sequence is called **divergent** if it is not convergent.

Example 4.2 — a convergent sequence

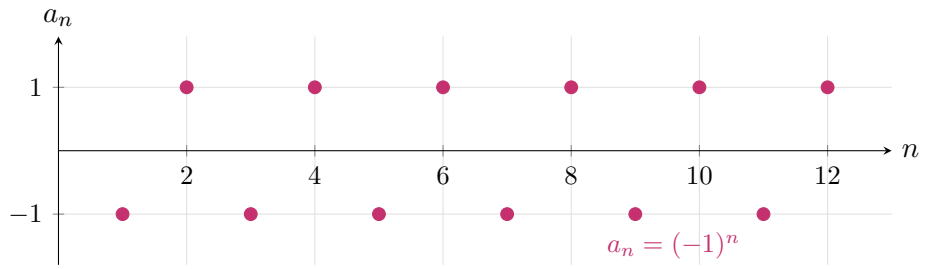
$\frac{1}{n}$ is convergent, with limit 0.

As n becomes very large, the numbers $\frac{1}{n}$ become very small — positive, and small — so they come closer and closer to 0.

Example 4.3 — an oscillating divergent sequence

$(-1)^n$ is divergent.

The terms are $-1, 1, -1, 1, -1, 1, \ldots$: the sequence **oscillates** and clearly does not come close to any real number, so it has no limit.



Example 4.4 — divergence in a different way

The sequences $a_n = n$ and $a_n = -n$ are divergent.

The sequence n is $1, 2, 3, 4, 5, 6, \ldots$. Take any real number — say two million — and at some stage the sequence goes beyond it and keeps going far beyond. It never comes close to any real number, so it has no limit.

Similarly $-n$ gives $-1, -2, -3, -4, \ldots$, becoming very small in the negative sense. Take any real number, say -1003.89: the terms -1000, -1001, -1002, -1003, -1004 have already gone past it, and the sequence keeps going smaller.

Two flavours of divergence

If you have met the concept of **infinity** — something larger than any real number we know — you may say the sequence n *converges to infinity* or *diverges to infinity*; likewise $-n$ converges or diverges to $-\infty$. Both phrases say more than merely "divergent", and as far as our terminology goes these sequences are still **divergent**.
But note the difference from $(-1)^n$: that one just jumps around and does not converge in any sense, whereas n and $-n$ really do go off somewhere — beyond our universe, if you think of the real line as the universe.

6 Subsequences

Definition 4.2 — Subsequence

A **subsequence** of a sequence is a new sequence formed by possibly excluding some entries of the original sequence.

Given $a_1, a_2, a_3, a_4, \ldots$ you may drop some terms and keep, say, $a_1, a_4, a_{20}, a_{100}, a_{1000000}, \ldots$ in some prescribed order — whatever you get is a subsequence. Often one drops terms in a mathematically interesting way: take the **odd part** (first, third, fifth, …) or the **even part** (second, fourth, sixth, …).

Example 4.5

Take the divergent sequence $(-1)^n$, i.e. $-1, 1, -1, 1, \ldots$, and keep the even terms.

The even terms are $(-1)^2, (-1)^4, (-1)^6, \ldots$, each of which equals 1, so the subsequence is

$$1,\ 1,\ 1,\ 1,\ \ldots$$

One could instead take the fourth, eighth, twelfth terms and so on — that is $(-1)^{4n}$ — which gives exactly the same numbers. As *sequences* these two are the same, but as *subsequences of the original* they are different, since one is built from the second, fourth, sixth, … terms and the other from the fourth, eighth, twelfth, … terms.

7 Sequences behind the exponential

Here is a series turned into a sequence: let

$$a_n = \sum_{k=0}^{n} \frac{x^k}{k!}.$$

So the second term of the sequence is

$$\frac{x^0}{0!} + \frac{x^1}{1!}.$$

As $n \to \infty$ this becomes

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x,$$

and this is in fact *how* e^x is defined. One may object: if we know e, why not just raise it to a power? But then — what is e? And e is itself defined this way, by taking $x = 1$:

$$e = \sum_{k=0}^{\infty} \frac{1}{k!},$$

after which one works out its value, as Euler did — one of the reasons the number is named after him.

Sequences converging to e and e^x — treat these as black boxes

$$\sum_{k=0}^{n} \frac{x^k}{k!} \longrightarrow e^x \qquad \left(1+\frac{x}{n}\right)^n \longrightarrow e^x$$

$$\left(\frac{n^n\sqrt{2\pi n}}{n!}\right)^{1/n} \longrightarrow e \qquad \frac{n}{\sqrt[n]{n!}} \longrightarrow e$$

For $\left(1+\frac{x}{n}\right)^n$: at $n=1$ the term is $(1+x)^1$, at $n=2$ it is $\left(1+\frac{x}{2}\right)^2$, and so on — and the sequence converges to e^x. The third expression comes from **Stirling's formula**, which you may meet in statistics in proofs concerning the normal distribution. Do not try to work out *why* these limits are what they are; take them as black boxes.

Why this matters

Behind every exponential lies this notion of sequences and series. The exponential turns up in mathematics, statistics, science and technology everywhere, and at its very heart is the idea of a sequence — we simply know how to handle it, so we do not keep repeating it. What you need to keep in mind is that there *are* such sequences whose limit is e or the exponential function.

8 Rules for convergence

We cannot keep evaluating terms and plotting dots — the third example moved so slowly that no amount of plotting would convince us it converges to 2. So we need theoretical tools: a stock of sequences whose limits we know, plus rules that let us build new limits from old.

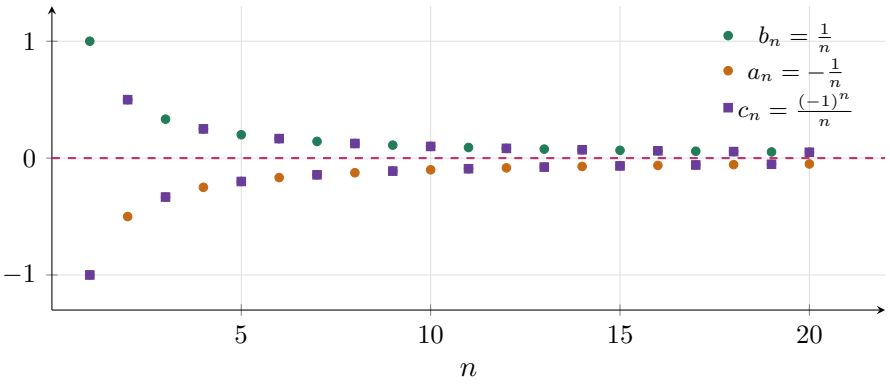
Ten rules

Let $a_n \to a$ and $b_n \to b$.

1. Every **subsequence** of a_n also converges to a.
2. $a_n + b_n \to a + b$.
3. $c\,a_n \to c\,a$ for $c \in \mathbb{R}$.
4. $a_n - b_n \to a - b$.
5. $a_n b_n \to a\,b$.
6. If f is a **polynomial** function in one variable, then $f(a_n) \to f(a)$.
7. If $b \neq 0$, then $\dfrac{a_n}{b_n} \to \dfrac{a}{b}$.
8. $c^{a_n} \to c^a$ for $c > 0$.
9. If $a_n > 0$ for all n, and $a > 0$, $c > 0$, $c \neq 1$ then $\log_c a_n \to \log_c a$.
10. **Sandwich principle:** if $a_n \to a$ and $b_n \to a$, and c_n satisfies $a_n \leq c_n \leq b_n$, then $c_n \to a$.

Rules 4 and 6 can in fact be derived from the others. These results do have proofs — given in a first course on calculus — which we do not do here. The last rule is called the **sandwich principle**

precisely because c_n is sandwiched between a_n and b_n: if the two outer sequences tend to the same thing, the middle one is forced to follow.



9 Applying the rules

Example 4.6

Show that

$$\frac{(-1)^n}{n} \longrightarrow 0.$$

The first few terms are

$$-1, \quad \frac{1}{2}, \quad -\frac{1}{3}, \quad \frac{1}{4}, \quad -\frac{1}{5}, \quad \dots$$

so the terms oscillate about 0 but come closer and closer to it. To prove it, apply the **sandwich principle**. We know $\frac{1}{n} \to 0$; multiplying by the constant -1 (Rule 3) gives $-\frac{1}{n} \to 0$. Now set

$$a_n = -\frac{1}{n}, \qquad c_n = \frac{(-1)^n}{n}, \qquad b_n = \frac{1}{n},$$

so that $a_n \le c_n \le b_n$ for every n. Since both outer sequences tend to 0,

$$\boxed{\frac{(-1)^n}{n} \longrightarrow 0}$$

(There are many ways of doing this; that is one of them.)

Example 4.7

Find the limit of

$$c_n = \frac{\dfrac{1}{\log(1+n)} + \dfrac{5n^2}{1+n^2}}{\left(1+\dfrac{1}{n}\right)^{2n}}.$$

This one is slightly harder, since it is not answered directly by the rules — so let us see what is going on.

The numerator

First, $\log(1+n)$: as $n \to \infty$ it moves very slowly, but it does go to infinity, so this sequence **diverges to infinity**. Consequently its reciprocal tends to 0:

$$\frac{1}{\log(1+n)} \longrightarrow 0.$$

Next, $\dfrac{5n^2}{1+n^2}$. Look at the numerator and denominator separately by dividing through by n^2:

$$\frac{5n^2}{1+n^2} = \frac{5}{\dfrac{1}{n^2}+1} \longrightarrow \frac{5}{1} = 5,$$

since $\frac{1}{n^2} + 1 \to 1$. So, calling the two pieces $a_n \to 0$ and $b_n \to 5$, the sum rule gives

$$a_n + b_n \longrightarrow 0 + 5 = 5.$$

The denominator

The denominator looks complicated, but it can be rewritten:

$$\left(1+\frac{1}{n}\right)^{2n} = \left[\left(1+\frac{1}{n}\right)^{n}\right]^2.$$

The term inside is exactly one of our known sequences, converging to e. Hence the denominator tends to e^2.

Putting it together

The denominator's limit e^2 is non-zero, so by the quotient rule

$$\boxed{c_n \longrightarrow \frac{5}{e^2}}$$

This is how the rules and the known examples are applied to compute actual limits.

Lecture 5 — Limits for Functions of One Variable

Left and right limits; limits at infinity; substitution and its dangers; continuity at a point

1 From sequences to functions

We have seen limits for a sequence of real numbers. Let us now build on that idea and talk about limits for a **function**.

Recall one of the rules: if $a_n \to a$ and $b_n \to b$ then $a_n b_n \to ab$; in particular

$$a_n \longrightarrow a \qquad \Longrightarrow \qquad a_n^2 \longrightarrow a^2.$$

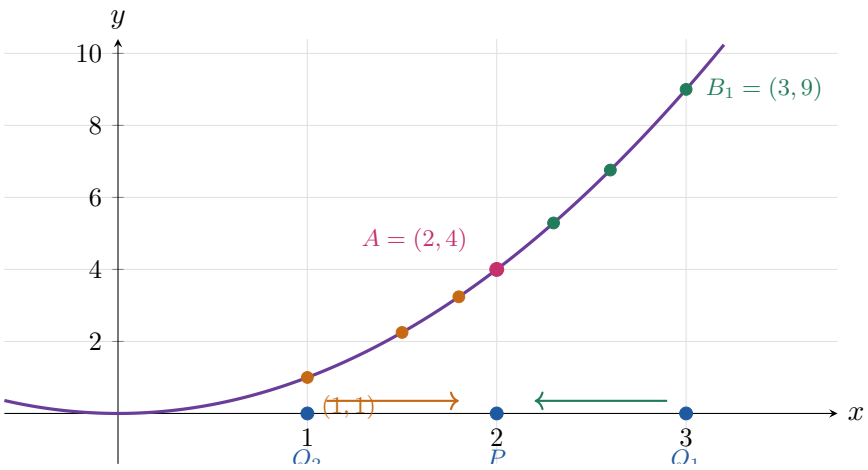
We can read this differently. Put $f(x) = x^2$. Then $f(a_n) = a_n^2$ and $f(a) = a^2$, so the statement becomes

$$\boxed{f(a_n) \longrightarrow f(a) \quad \text{whenever} \quad a_n \longrightarrow a}$$

This is an equivalent formulation, and it is really a statement about the **limit of f at a**. The question of this lecture is: as a sequence of points approaches a particular point, how does the corresponding sequence of *function values* behave?

1.1 The picture for $f(x) = x^2$ at $a = 2$

Take $a = 2$, so $P = (2, 0)$ and the corresponding point on the graph is $A = \big(2, f(2)\big) = (2, 4)$.



From the right: start at $Q_1 = (3, 0)$, whose function value gives $B_1 = (3, 9)$. As the sequence moves towards P, the point on the graph moves quickly towards A.

From the left: start at $Q_2 = (1, 0)$, giving the graph point $(1, 1)$. Watching the values, the inputs creep up as $1.93, 1.99, 1.999, \ldots$ while the function values run $3.7249, 3.9601, 3.996001, \ldots$ — coming closer and closer to 4. Carry on long enough (the animation was stopped at $n = 50{,}000$) and the values come very close indeed to 4.

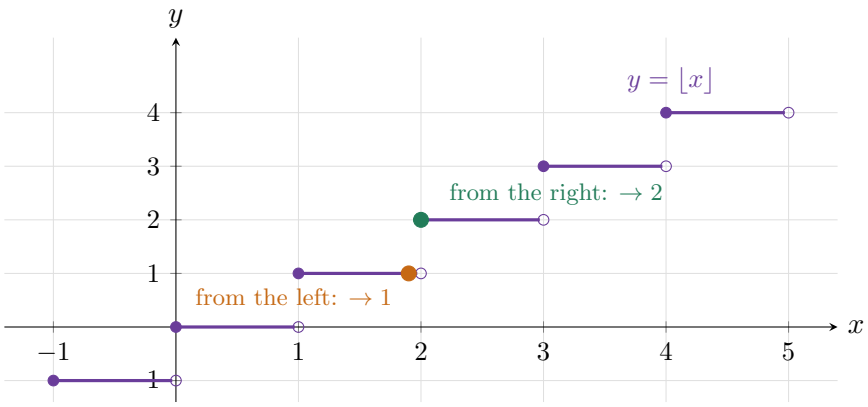
So from both sides, as the sequence approaches 2, its function value approaches $f(2)$. This is exactly the statement $f(a_n) \to f(a)$.

2 Contrast: the floor function

Is this a general phenomenon? Let us contrast $f(x) = x^2$ with the floor function $g(x) = \lfloor x \rfloor$ — the largest integer less than or equal to the number. For instance

$$\lfloor -6.5 \rfloor = -7,$$

so it is the integer part for positive numbers and one below the integer part for negative ones.



Between two integers the function takes the same value: $\lfloor 3.5 \rfloor = 3$, $\lfloor 3.98 \rfloor = 3$; but at 4 it jumps to 4. At every integer there is a jump, and after that it is constant — a step function.

- **Approaching 2 from the right**, using for instance the sequence $a_n = 2 + \dfrac{1}{(1+\log n)^{1.1}}$, we find $g(a_n) \to g(2) = 2$. Everything works.
- **Approaching 2 from the left**, the terms are 1.94733, 1.95086, 1.95216, ... — they never actually reach 2, so after some point they always lie strictly between 1 and 2, and therefore

  $$g(a_n) = 1 \qquad \text{always.}$$

  So $g(a_n) \to 1$, which is *not* $g(2) = 2$.

So the phenomenon $f(a_n) \to f(a)$ is **special**. Looking at the graphs, the difference is plain: the floor function has a **jump**, whereas x^2 is a very nice function. The limit is precisely the tool that picks up such breaks — and that is why we are studying limits. (Recall that our study of limits began right after the video on tangents, because a tangent requires understanding how a curve behaves as we come close to a point.)

At non-integer points there is no problem

For g this failure happens at each **integer** value. If a is a non-integer — say $a = 1.5$ — then from both sides the function value is 1, so $g(a_n) \to g(a)$ perfectly well. All the jumps sit at the integers.

3 A pathological example

Such cases matter because they stop us making false statements: whenever you want to claim something about limits, run these examples past it first.

Example 5.1 — a function with jumps everywhere

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \quad \text{(rational)}, \\ 0, & x \notin \mathbb{Q} \quad \text{(irrational)}. \end{cases}$$

There is no good expression for this function, and you simply cannot draw it — that is rather the point. It has jumps **everywhere**, so we should expect trouble with limits, and indeed there is.

Recall that $\sqrt{2}$ is irrational — it cannot be written as $\frac{m}{n}$ with m, n integers — so

$$f\big(\sqrt{2}\big) = 0.$$

Now construct a sequence of *rational* numbers tending to $\sqrt{2}$. Since $\sqrt{2} = 1.414\ldots$, take

$$a_1 = 1.4 = \frac{14}{10}, \qquad a_2 = 1.41 = \frac{141}{100}, \qquad a_3 = 1.414 = \frac{1414}{1000}, \qquad \ldots$$

In general, let a_n be the number obtained by taking the first n digits after the decimal point in $\sqrt{2}$. Each has a finite decimal expansion, hence is rational, and clearly $a_n \to \sqrt{2}$.

But then $f(a_n) = 1$ for every n, since each a_n is rational, so

$$f(a_n) \longrightarrow 1 \neq 0 = f\big(\sqrt{2}\big).$$

So even though $a_n \to \sqrt{2}$, we do **not** have $f(a_n) \to f\big(\sqrt{2}\big)$.

Try it yourself

There is nothing special about $\sqrt{2}$ — the same works at any point. Pick your favourite irrational number, say $\sqrt{3}$ or $\sqrt{5}$, and check that the same thing happens. Slightly harder: construct a sequence of *irrational* numbers tending to a *rational* number, say to 1 or to 0.

4 Definitions: left, right and two-sided limits

We need these definitions precisely because of what happened for the step function.

Definition 5.1 — Limit from the left

Let f be a function and a a point such that $a_n \to a$ for some sequence a_n in the domain of f. If there is a real number L such that

$$f(a_n) \longrightarrow L \quad \text{for } \textbf{all} \text{ sequences } a_n \to a \text{ with } a_n < a,$$

then we say the limit of f at a **from the left** exists and equals L, written

$$\lim_{x \to a^-} f(x) = L.$$

If there is no such L, the limit from the left **does not exist**.

Definition 5.2 — Limit from the right

Similarly, if there is a real number R such that $f(a_n) \to R$ for **all** sequences $a_n \to a$ with $a_n > a$, then the limit of f at a **from the right** exists and equals R, written

$$\lim_{x \to a^+} f(x) = R.$$

The equivalent way to think of it

$\lim_{x\to a^-} f(x) = L$ means: as x comes closer and closer to a **from the left**, $f(x)$ eventually comes closer and closer to L. Likewise for the right.
Keep this equivalence in mind — it lets us stop talking about sequences altogether, which is what we shall do from the next lecture onwards.

Definition 5.3 — Limit of a function at a point

Suppose the limits of f at a from both sides exist and are equal:

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x), \qquad \text{i.e.} \quad L = R.$$

Then we say the limit of f at a exists and equals that common value, written

$$\lim_{x\to a} f(x).$$

If either one-sided limit fails to exist, or both exist but are unequal, we say the limit **does not exist** (DNE).

a need not be in the domain

Note that in the one-sided definitions we assumed nothing about a belonging to the domain of f — f need not be defined at a at all. All we need is a sequence $a_n \to a$ on which f is defined. We shall see examples of exactly this.

4.1 The three examples revisited

Function	Left limit	Right limit	Limit
$f(x) = x^2$	a^2	a^2	$a^2 = f(a)$
$g(x) = \lfloor x \rfloor$, a not an integer	$\lfloor a \rfloor$	$\lfloor a \rfloor$	$\lfloor a \rfloor = g(a)$
$g(x) = \lfloor x \rfloor$, $a = 2$	1	2	DNE
$f = 1$ on $\mathbb{Q}$, 0 off $\mathbb{Q}$	DNE	DNE	DNE

For x^2 both one-sided limits exist and match, and moreover equal $f(a)$ — a very special function. For the floor function at $a = 2$ both one-sided limits exist but do not match, so $\lim_{x\to 2} g(x)$ does not exist. For the rational-indicator function neither one-sided limit exists, so the limit does not exist either.

5 Limits at infinity

Definition 5.4 — Limits as $x \to \pm\infty$

Let f be defined for all $x > M$ for some M. If for **all** sequences x_n diverging to infinity there is an L with $f(x_n) \to L$, then

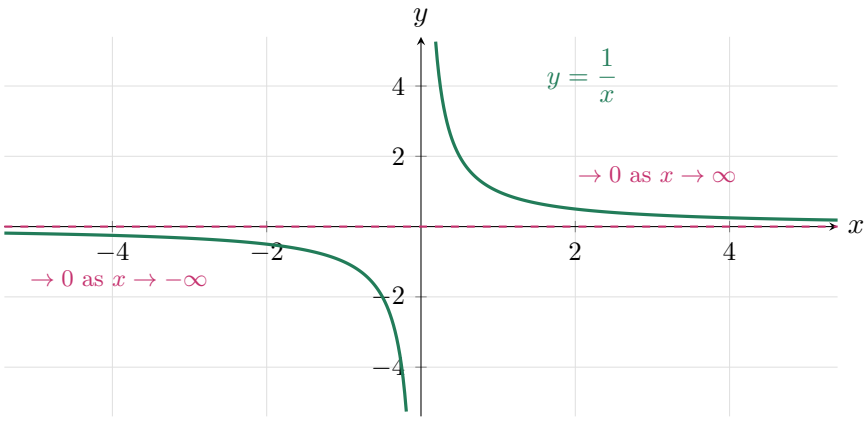
$$\lim_{x\to\infty} f(x) = L.$$

Similarly, if f is defined for all $x < N$ for some N, and for all sequences x_n diverging to $-\infty$ we have $f(x_n) \to L$, then $\lim_{x\to-\infty} f(x) = L$.

We need f to be defined for all large numbers, otherwise we cannot speak of $x \to \infty$ at all. (By "small" on the negative side we do not mean close to 0 — we mean large negatives.)

Example 5.2

Study $f(x) = \dfrac{1}{x}$.



As x increases along the axis the curve comes closer and closer to the axis, so the function value approaches 0; the same happens for large negatives (the values are negative, but very close to 0). Hence

$$\lim_{x\to\infty} \frac{1}{x} = 0, \qquad \lim_{x\to-\infty} \frac{1}{x} = 0.$$

The real trouble is at 0:

$$\lim_{x\to0^+} \frac{1}{x} \quad \text{and} \quad \lim_{x\to0^-} \frac{1}{x} \quad \text{both do not exist.}$$

On one side the values diverge to $+\infty$ and on the other to $-\infty$; they do not converge, and even if you allow infinities into your world, the two do not match.

Why this matters

Limits at $\pm\infty$ are important in many real-life phenomena. You will meet the exponential and normal distributions and talk about the **tails** of distributions, where exactly this question — what happens as $x \to \infty$ or $x \to -\infty$ — is what is being asked.

6 Some basic limits

Let k be an integer. Then:

$$\lim_{x\to a} x^k = a^k \quad (k > 0) \qquad \lim_{x\to a} x^k = a^k \quad (k < 0,\ a \neq 0)$$

$$\lim_{x\to a} e^x = e^a \qquad \lim_{x\to a} \log x = \log a \quad (a > 0)$$

$$\lim_{x\to a} \sin x = \sin a \qquad \lim_{x\to a} \tan x = \tan a \quad \left(a \neq (2n+1)\tfrac{\pi}{2}\right)$$

For all these nice functions we simply substitute the value of a into the function. And even where the limit fails — for instance $k < 0$ with $a = 0$, the $\frac{1}{x}$ situation — it is clear from the graph that it does so.

7 Beware of substitution

★ Beware ★

Often we can substitute a into the expression for $f(x)$ and obtain the limit. But this **does not work** when the function gets slightly complicated, or when a does not belong to the domain of definition of f — and remember, a need not belong to it.

Example 5.3

$$\lim_{x\to 2}\frac{x^2-5x+6}{x-2} \qquad \text{and} \qquad \lim_{x\to 0}\frac{1}{x}.$$

For the second: substituting $x=0$ obviously makes no sense — this is our good old friend, division by zero — and here the limit genuinely **does not exist**.

For the first, let us substitute and see:

$$\frac{2^2-5\cdot 2+6}{2-2}=\frac{4-10+6}{0}=\frac{0}{0}.$$

Warning bells. A $\frac{0}{0}$ situation is one of the most interesting: unlike the case on the right, where we can see at once that the limit does not exist, here we cannot substitute — but the limit *may* still exist. And indeed it does, because the expression factorises:

$$\frac{x^2-5x+6}{x-2}=\frac{(x-2)(x-3)}{x-2}=x-3.$$

So the function is the same as $x-3$ — for all values *except* $x=2$, where the original expression is not defined at all, while $x-3$ is defined and equals -1. Hence as x approaches 2,

$$\boxed{\lim_{x\to 2}\frac{x^2-5x+6}{x-2}=-1}$$

The moral

Substitution does work for nice, special functions — $\sin x$, $\tan x$, e^x, polynomials, and so on. But in general, even for **rational functions** (polynomial divided by polynomial), you must be very careful about what is happening.

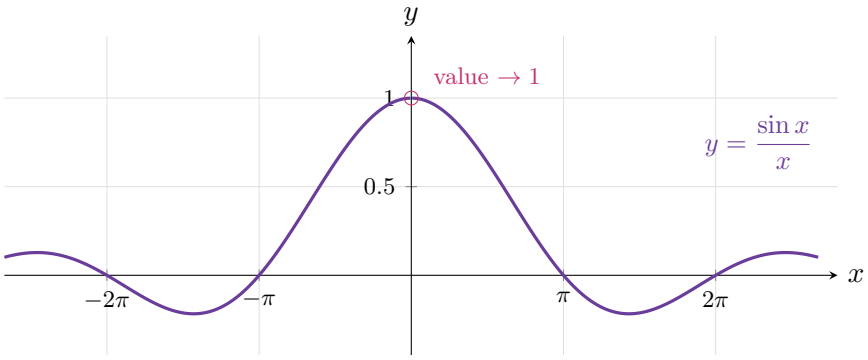
8 Three important limits

$$\lim_{x\to 0}\frac{\sin x}{x}=1, \qquad \lim_{x\to 0}\frac{\ln(1+x)}{x}=1, \qquad \lim_{x\to\infty}\frac{a+b\,e^x}{c+d\,e^x}=\frac{b}{d}$$

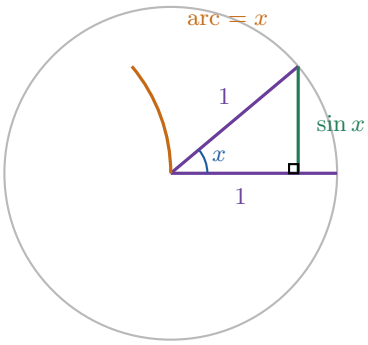
Note that the first two are again $\frac{0}{0}$ situations on substitution — $\sin 0=0$ over 0 — so substitution makes no sense. Unlike the earlier example, there is no easy factorisation to rescue us; we need to know something more about these functions.

8.1 Why $\lim_{x\to 0} \frac{\sin x}{x} = 1$

Numerically. Take the sequence $x = \frac{1}{n} \to 0$ and watch $\dfrac{\sin(1/n)}{1/n}$: it becomes 1 almost at once and then stops changing. Take instead the far slower sequence $x = -\dfrac{1}{(1+\log m)^{1.1}} \to 0$ and the values run 0.999, 0.99946, 0.99947, …, ending around 0.9995 — and continuing to close in on 1.



Geometrically. Work in radians and take the **unit circle**. Let the arc subtend an angle of x radians, so the arc length is x and the two radii have length 1. Dropping a perpendicular gives a right triangle with hypotenuse 1, so the opposite side has length $\sin x$.



From this picture $\dfrac{\sin x}{x}$ is clearly less than 1; one can also bound it from the other side by a quantity tending to 1. Then both bounds go to 1, so the middle quantity goes to 1 — exactly the **sandwich principle** we met for sequences.

8.2 The logarithmic limit

For $\dfrac{\log(1+x)}{x}$ there is likewise no direct computation available. The proof is done more formally in the next lecture, and rests on the inequalities bounding it on one side by 1 and on the other by $\dfrac{1}{1+x}$.

8.3 The exponential limit

Here we can give a justification. Substituting $x = \infty$ gives infinity over infinity, which tells us nothing. Instead multiply numerator and denominator by e^{-x}:

$$\frac{a + b\,e^{x}}{c + d\,e^{x}} = \frac{a\,e^{-x} + b}{c\,e^{-x} + d}.$$

Now we may take the limit, because e^{-x} behaves much like $\frac{1}{x}$ — it goes to 0 very fast:

$$\lim_{x\to\infty} \frac{a\,e^{-x}+b}{c\,e^{-x}+d} = \frac{a\cdot 0+b}{c\cdot 0+d} = \boxed{\frac{b}{d}}$$

So it *is* possible to substitute sometimes, but not always — and for a particular expression you may not be able to substitute, yet after rewriting the expression you can. Much of understanding limits is understanding exactly when this is allowed.

9 Continuity at a point

Definition 5.5 — Continuity at a point

Let f be a function and a a point such that $a_n \to a$, where a_n **and** a **both belong to the domain of** f. Then f is said to be **continuous at** a if

$$\lim_{x\to a} f(x) = f(a).$$

Continuity means the limit at a can be obtained by **evaluating** the function at a.

Evaluating, not substituting

The word is deliberately *evaluating* rather than *substituting*. Substituting means putting a into a given expression, which may or may not be possible. Evaluating may still make sense — sometimes because you can rewrite the expression into an effectively identical function and then evaluate; and sometimes because the function has no expression at all, yet can still be evaluated.

Example 5.4

Define

$$f(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

The expression $\frac{\sin x}{x}$ makes no sense at 0, so we *define* the value there to be 1. Then f is **continuous at** 0, since the limit as $x \to 0$ is 1, which is exactly the value obtained by evaluating f at 0.

To recap the structure of the definitions: for the **left limit** we require that for *all* sequences approaching from the left — not a few, not some particular ones, all of them — the function values tend to a common number L; similarly from the right for a number R; if L and R match, the limit exists; and if it further equals the value of the function at that point, the function is continuous there.

References and Acknowledgement

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Notes created by: Vishal (Instructor)